

Fig. 27: Frequency range varied

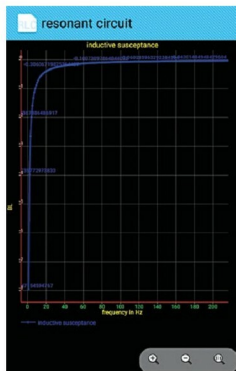


Fig. 28: Inductive susceptance versus frequency

all parallel RLC circuit is shown in Fig. 22.

In this RLC Android application, you can simulate inductive and capacitive susceptances, admittance, current, phase angle, bandwidth, inductive and capacitive currents graphs.

To view the graph clearly, manually adjust X-axis range. By default,

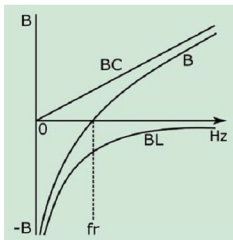


Fig. 29: Susceptance versus frequency

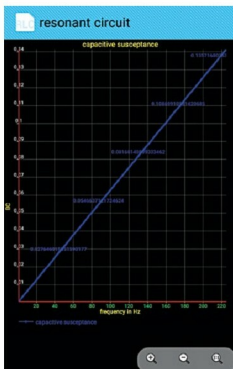


Fig. 30: Capacitive susceptance versus frequency

axis labels will be 15 (Fig. 24). If you want to view more XY labels, increase the axis label count.

Parallel RLC circuit at resonance. In the parallel resonance circuit, inductive reactance X_L and capacitive reactance X_C is the same as in the series resonance circuit. So no details on reactances are discussed here.

The following graph plots are based on the inputs as shown in Fig. 21.

Admittance (Y). In Fig. 25, admittance is the inverse of impedance and is denoted by Y and its unit is mho or siemens. At resonance, admittance is equal to conductance G , that is, reciprocal of resistance.

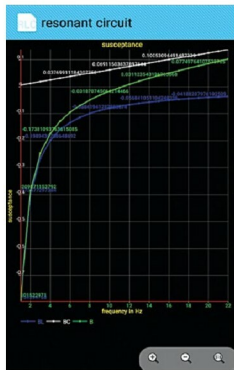


Fig. 31: Susceptance versus frequency plot

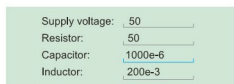


Fig. 32: Modified input

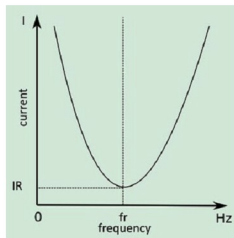


Fig. 33: Ideal case of current versus frequency

In Fig. 26, at resonance, admittance value ($Y = 0.02$ s) is equal to conductance value ($G = 0.02$ s) and this graph is simulated from 90Hz to 140Hz as shown in Fig. 27.

Inductive susceptance (BL). Inductive susceptance is denoted by BL and is negative in value. Inductive susceptance is inversely proportional to frequency. In Fig. 28, BL graph indicates the BL curve as shown in the